

SPECIFICATION
FOR
AMOLED+Lens Module

MODULE No:	KD070FH0IN124-C142A
CUSTOMER:	

STARTEK	INITIAL	DATE
PREPARED BY		
CHECKED BY		
APPROVED BY		

CUSTOMER	INITIAL	DATE
APPROVED BY		



Revision History

[illegible]

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常 备 库 存 Stock For Sale	长 期 供 货 Long Time supply	支持小量 NO MOQ	品 种 齐 全 In Full Range	

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1.Basic Specifications

* Description

This is a color active matrix AMOLED module using Low Temperature Poly-silicon Thin Film Transistors as a ctive switching devices. This module has a 7 inch diagonally measured active area with 1080 horizontal by 1920 vertical pixel arrays. Each pixel is divided into RED and GREEN dots, or BLUE and GREEN dots, and two pixels share RED or BLUE dots which are arranged in vertical stripe and this module can display 1073.7M colors.

1.1 OLED Features

General Information	Items	Specification	Unit	Note
		Main Panel		
	Display area(AA)	86.94(H)*154.56(V) (7.0 inch)	mm	
	Display mode	AMOLED	-	
	Display colors	1073.7M	colors	
	Number of pixels	1080(RGB)*1920	dots	
	Pixel arrangement	RGB vertical stripe	-	
	Pixel pitch	0.0805(H)*0.0805(V)	mm	
	Viewing angle	ALL	o'clock	
	Controller IC	GD7658A V110	-	
	Touch Controller IC	Focal3519(BGA)	-	
	LCM Interface	MIPI(+SPI)	-	
	Operating temperature	-30~+85	℃	
	Storage temperature	-40~+85	℃	
	Module bonding technology	Use Optical bonding between LCM and LENS	-	

1.2 LENS Features

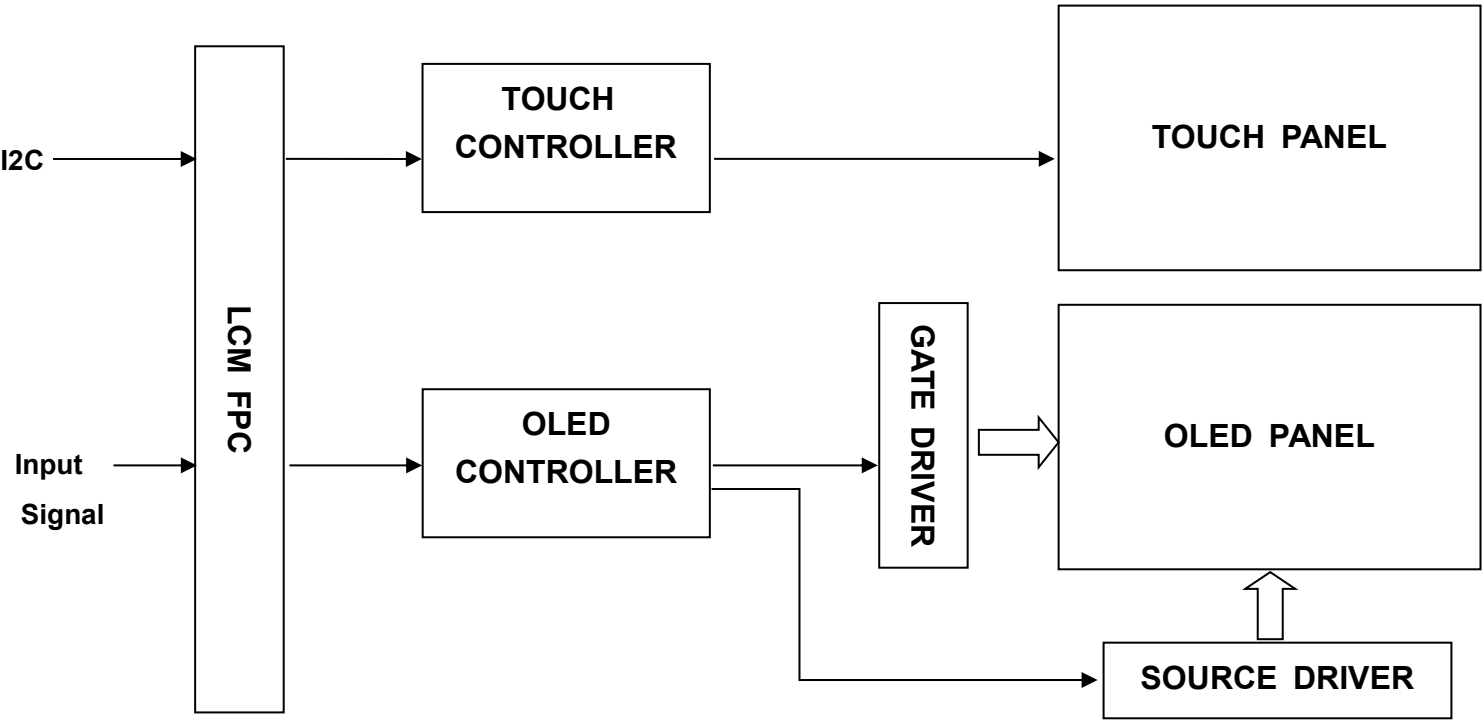
General Information	Items	Specification	Unit	Note
		Main Panel		
	Material	Glass	-	
	Surface hardness	≥6H	-	
	Surface treatment	NA	-	

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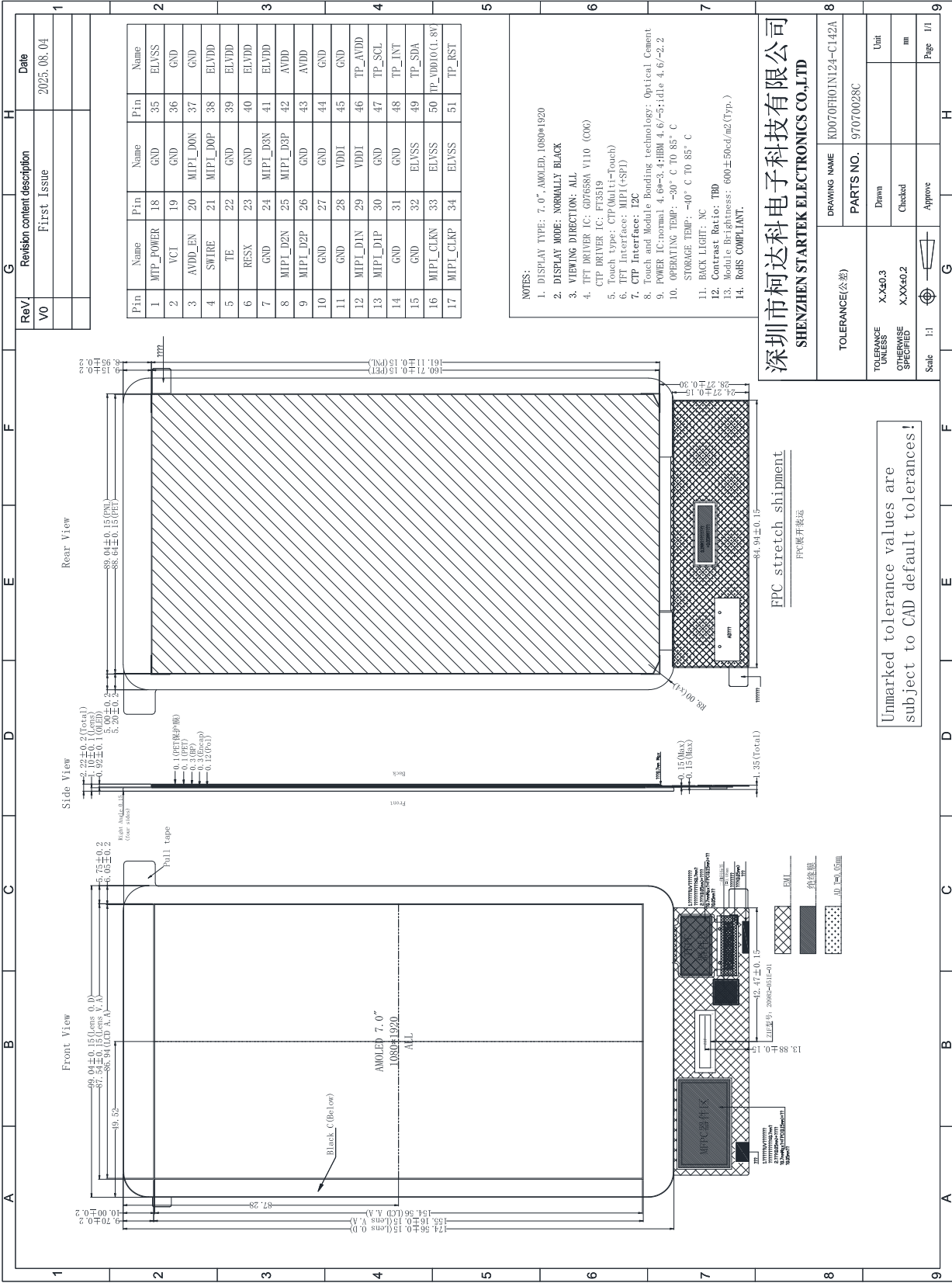
1.3 Mechanical Information

Item		Min.	Typ.	Max.	Unit	Note
Module size	Horizontal(H)	-	99.04	-	mm	
	Vertical(V)	-	174.56	-	mm	
	Depth(D)	-	2.22	-	mm	
Weight		-	TBD	-	g	

2. Block Diagram



3. Outline dimension



4. Input terminal Pin Assignment

(Connector:20982-051E-01)

NO.	SYMBOL	DISCRIPTION	I/O
1	MTP_POWER	Power supply for OTP. Float it for normal operation.	I
2	VCI	Analog Power for Driver IC	P
3	AVDD_EN	Enable ELVDD and ELVSS output of DC/DC IC	I
4	SWIRE	Enable ELVDD and ELVSS output of DC/DC IC	I
5	TE	Feed Back From Driver IC for synchronism	I
6	RESX	Reset pin for DDIC,active low.	I
7	GND	Ground	P
8	D2N	MIPI DSI data 2-for Master IC	I
9	D2P	MIPI DSI data 2-for Master IC	I
10	GND	Ground	P
11	GND	Ground	I
12	D1N	MIPI DSI data 1-for Master IC	I
13	D1P	MIPI DSI data 1-for Master IC	P
14	GND	Ground	I
15	GND	Ground	I
16	CLKN	MIPI DSI Clock-for Master IC	P
17	CLKP	MIPI DSI Clock-for Master IC	I
18	GND	Ground	I
19	GND	Ground	P
20	D0N	MIPI DSI data 0-for Master IC	I
21	D0P	MIPI DSI data 0-for Master IC	I
22	GND	Ground	P
23	GND	Ground	P
24	D3N	MIPI DSI data 3-for Master IC	I
25	D3P	MIPI DSI data 3-for Master IC	I
26	GND	Ground	P
27	GND	Ground	P
28	VDDI	Digital power for Driver IC	P
29	VDDI	Digital power for Driver IC	P
30	GND	Ground	P

31	GND	Ground	P
32	ELVSS	ELVSS for OLED	P
33	ELVSS	ELVSS for OLED	P
34	ELVSS	ELVSS for OLED	P
35	ELVSS	ELVSS for OLED	P
36	GND	Ground	P
37	GND	Ground	P
38	ELVDD	ELVDD for OLED	P
39	ELVDD	ELVDD for OLED	P
40	ELVDD	ELVDD for OLED	P
41	ELVDD	ELVDD for OLED	P
42	AVDD	Charge pumping Power for Driver IC	P
43	AVDD	Charge pumping Power for Driver IC	P
44	GND	Ground	P
45	GND	Ground	P
46	TP_AVDD	Analog Power Supply(TYP.3.3V)	P
47	TP_SCL	TP Slave clock for SPI(Serial Peripheral Interface)(1.8V)	I
48	TP_INT	TP INT for AP(1.8V)	I
49	TP_SDA	TP Master output	I
50	TP_1V8	Analog Power Supply(TYP.1.8V)	P
51	TP_RST	TP Reset(1.8V)	I

5. OLED Optical Characteristics

5.1 Optical specification

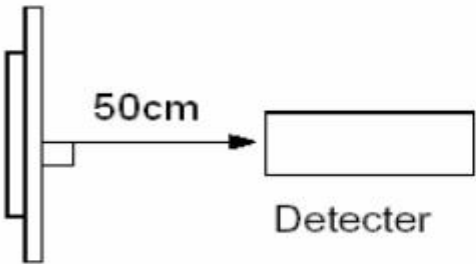
Item		Symbol	Condition	Min.	Typ.	Max.	Unit.	Note
Contrast Ratio		CR	Θ=0 Normal viewing angle	100000	--	--		3.4
Response time		Tr		--	--	3	msec	3.5
		Tf						
		Tgray						
Luminance (with pol)		Normal mode Bp	Θ=0° ϕ=0°	480	540	--	cd/m2	CPK>1.3 3
		HBM mode Bp	Θ=0° ϕ=0°	800	900	--	cd/m2	
		IDLE mode Bp			50		cd/m2	
Uniformity		△Bp		80	95	--	%	3.2
Color temperature		CT		7000	7500	8000	K	
Color Gamut		NTSC	Ratio	95	100	--	%	3.6
Color Filter Chromaticity	White	W _X	-0.04		0.3034	+0.04		(1)(4) CF glass
		W _Y			0.3232			
	Red	R _X			0.6890			
		R _Y			0.3109			
	Green	G _X			0.2604			
		G _Y			0.6625			
	Blue	B _X			0.1376			
		B _Y			0.0574			
Viewing angle	Hor.	ΘL	CR>10	85	--	--		3.3
		ΘR		85	--	--		
	Ver.	ΘU		85	--	--		
		ΘD		85	--	--		

Flicker	amount	Interval screen	--	--	-30	dB	
Gamma	Ddle mode		2.0	2.2	2.4		3.7(G16~G2 38)
	Normal mode		2.0	2.2	2.4		
	HBM mode		2.0	2.2	2.4		
Crosstalk	OLED Δ CT		--	--	2%		3.8
Transmittance (Without lens)	Tr	550nm	2%			%	
Color shift		Θ L=30°	--	--	4	JNCD	3.9
		Θ R=30°	--	--	4	JNCD	
		Θ U=30°	--	--	4	JNCD	
		Θ D=30°	--	--	4	JNCD	
OLED life time	LT50° (without lens 300 nit)	At 25℃,with white color pattern	--	20000	--	Hrs.	3.10
Option View Direction	ALL						

3.1: Luminance measurement

The test condition is at 25°C and measured on the surface of OLED module.

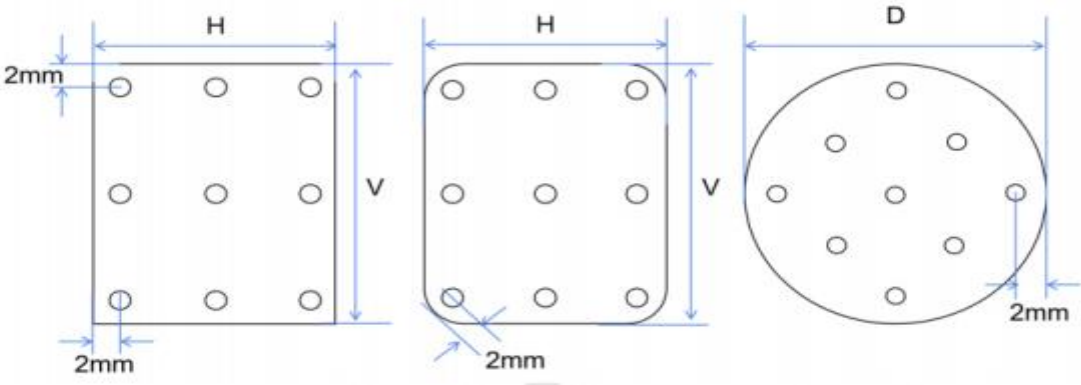
- The data are measured after OLEDs are lighted on for more than 5 minutes and displays
- equipment CS2000/CS2000A or similar equipments (Field of view:1deg,Distance:50cm)
- Measuring surroundings: Dark room.
- Adjust operating voltage to get optimum contrast at the center of the display.
- Measured value at the center point of panel must be after more than 5 minutes while light up



3.2: Uniformity

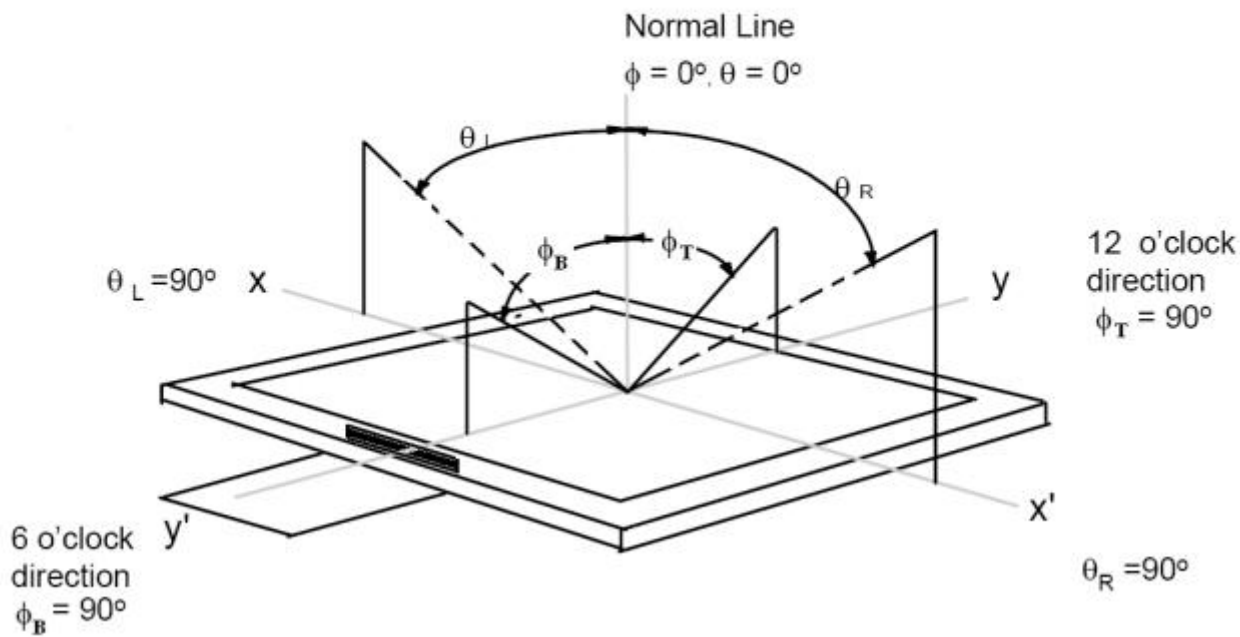
- The test condition is at 25°C and measured on the surface of display module
- Measurement equipment: CS2000/CS2000A or similar equipment.
- The luminance uniformity is calculated by using following formula.

$$\Delta Bp = Bp \text{ (Min.)} / Bp \text{ (Max.)} \times 100 \text{ (\%)}$$
- Bp (Max.) = Maximum brightness in 9 measured spots
- Bp (Min.) = Minimum brightness in 9 measured spots.



3.3: The definition of Viewing Angle

Refer to the graph below marked by θ and ϕ



3.4: The definition of Contrast Ratio (Test OLED using CS2000/CS2000A or similar equipments):

Contrast Ratio(CR)=

Luminance is at "White" state

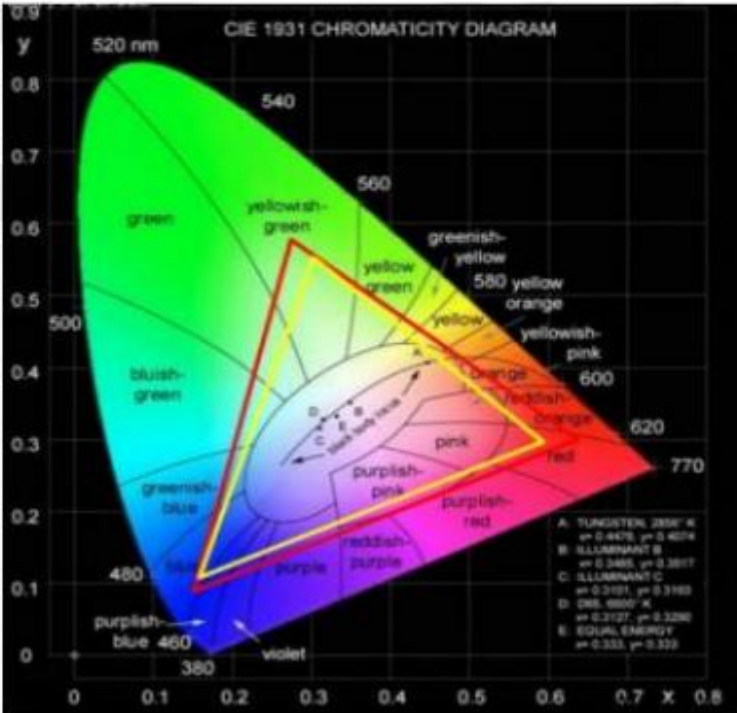
Luminance is at "Black" state

3.5: Definition of Response time.
TBD

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3.6: Definition of Color of CIE Coordinate and NTSC Ratio.

$$S = \frac{\text{area of RGB triangle}}{\text{area of NTSC triangle}} \times 100\%$$



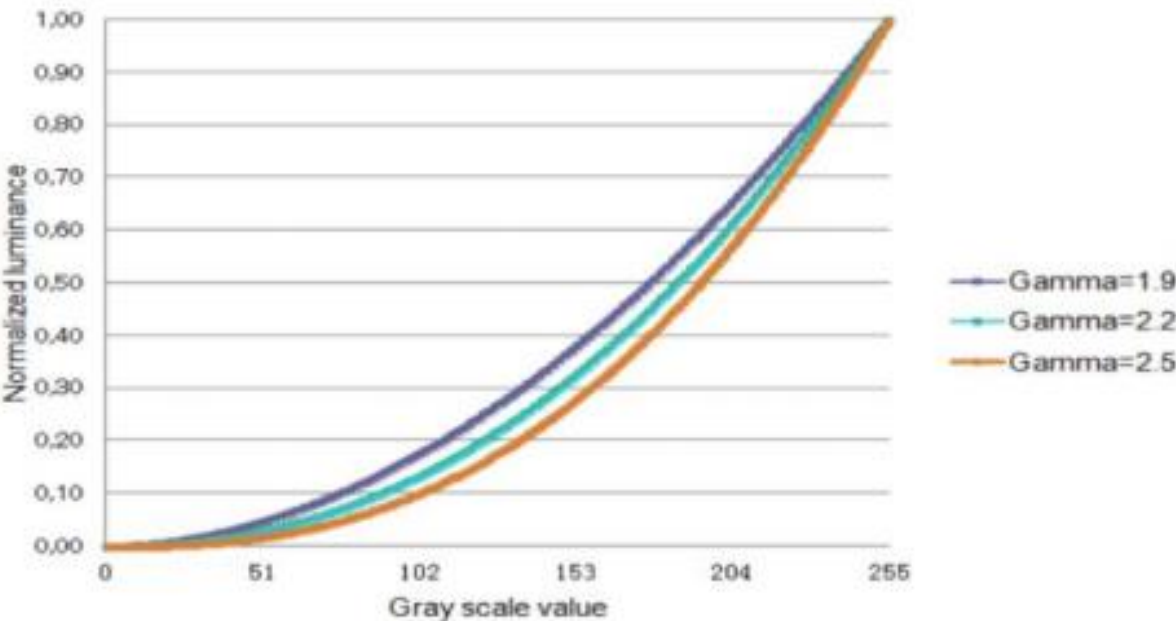
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3.7: gamma curve control

- For gamma curve control, request as below:
- the whole curve’s tolerance must control within +/-0.2, will test the gray scale below:
0~255gray, interval 8 gray (0,4,8,16.....255)
- According to below formula chart the curves

$$TR\%x=\sqrt{\frac{Gx-G0}{G255-G0}}$$

Gamma chart



3.8: Crosstalk

TBD

3.9: Color Shift JNCD

- For JNCD measure:
- Fix on one pattern like white pattern,
- On the condition $\theta=0$ $F=0^\circ$, we can get the color coordinate (u_1', v_1') and on $\theta_L=30^\circ$ we can get another color coordinate (u_2', v_2')
- $\Delta = \text{Square Root}((u_2' - u_1')^2 + (v_2' - v_1')^2)$
- JNCD stands for "Just Noticeable Color Difference"
- For the (u', v') color space $JNCD=0.0040$.

3.10 OLED lifetime

- Measurement equipment: CS2000/CS2000A or similar equipment.
- At room temperature(25°C), light the module with typical value brightness, display a white pattern,

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6. Electrical Characteristics

6.1 Power Consumption of Display Panel

Item		Symbol		Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
VCI		VCI		-	VCI	2.65	3	4.8	V	
VDDIO		VDDIO		-	VDDIO	1.65	1.8	1.95	V	
AVDD		AVDD		-	AVDD	6	7.6	8	V	
Power Consumption	Normal	Driver IC	VCI	100% Pixel On, 600nits, 120Hz	Ivci	-	0.35	0.42	mA	.
			VDDIO		Pvci	-	1.09	1.30	mW	
			AVDD		Ivddio	-	86.24	90.55	mA	
		Panel	ELVDD		Pvddio	-	159.54	167.52	mW	
			ELVSS		Iavdd	-	41.94	44.04	mA	
					Pavdd	-	318.74	334.68	mW	
		Panel	ELVDD		Ielvdd	-	412.25	432.86	mA	
			ELVSS		Pelvdd	-	1896.35	1991.17	mW	
					Ielvss	-	411.28	431.84	mA	
					Pelvss	-	1398.35	1468.27	mW	
	HBM	Driver IC	VCI	100% Pixel On, 1000nits, 120Hz	Ivci	-	0.35	0.41	mA	
			VDDIO		Pvci	-	1.07	1.28	mW	
			AVDD		Ivddio	-	90.00	94.50	mA	
		Panel	ELVDD		Pvddio	-	166.50	174.83	mW	
			ELVSS		Iavdd	-	44.74	46.98	mA	
					Pavdd	-	340.02	357.03	mW	
		Panel	ELVDD		Ielvdd	-	634.43	666.15	mA	
			ELVSS		Pelvdd	-	2918.38	3064.30	mW	
					Ielvss	-	676.00	709.80	mA	
					Pelvss	-	3380.00	3549.00	mW	
	AOD	Driver IC	VCI	100% Pixel On, 50nits, 30Hz	Ivci	-	0.28	0.33	mA	
			VDDIO		Pvci	-	0.86	1.03	mW	
			AVDD		Ivddio	-	46.75	49.08	mA	
		Panel	ELVDD		Pvddio	-	86.48	90.81	mW	
			ELVSS		Iavdd	-	9.17	9.63	mA	
					Pavdd	-	69.69	73.18	mW	

		Panel	ELVD		Ielvdd	-	33.95	35.64	mA	
			D		Pelvdd	-	156.16	163.97	mW	
			ELVSS		Ielvss	-	33.4	35.07	mA	
					Pelvss	-	73.47	77.15	mW	
Frame Rate		F _{frm}	HBM&NO	F _{frm}	-	120	-	Hz		
			R		-	30	-	Hz		
			AOD							

6.2 Absolute Maxmum Ratings

Item	Symbol	Min.	Max.	Unit	Note
Touch power supply voltage	VDD3-VSS	-0.3	3.6	V	
Touch Input voltage for I/O bus	IOVCC	-0.3	3.6	V	
Driver IC Operating temperature	Topr	-40	+85	°C	
Driver IC Storage temperature	Tstg	-55	+125	°C	
Touch IC Operating temperature	Topr	-30	+85	°C	
Touch IC Storage temperature	Tstg	-30	+85	°C	

7. MIPI-PHY Characteristics

7.1 DPHY LP/HS Characteristics

Signal	Symbol	Parameter	Min.	Typ.	Max.	Unit
HS_RX	V _{IDTH}	Differential input high threshold	-	-	70	mV
	V _{IDTL}	Differential input low threshold	-70			
	V _{IHHS}	Single-ended input high voltage			460	
	V _{ILHS}	Single-ended input low voltage	-40			
	V _{CMRX(DC)}	Common-mode voltage HS receive mode	70		330	
	Z _{ID}	Differential input impedance	-	85	-	Ω
LP_RX	V _{IL}	Logic0 voltage not in ULP State	-	-	550	mV
	V _{IH}	Logic1 input voltage	740	-	-	mV
	V _{LEAK}	I/O leakage current	-10	-	10	uA
LP_TX	V _{OL}	The venin output low level	-50		50	mV
	V _{OH}	The venin output high level	1.1	1.15	1.3	V

	Z _{OLP}	Output impedance of LP transmitter	110	-	-	Ω
--	------------------	------------------------------------	-----	---	---	---

7.2 MIPI DSI High-Speed RX Clock and Data-Clock Timing

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
F _{DSICLK}	DISCLK Frequency	VDDI=1.6 2~1.98V	40	-	650	MHz
T _{CLKP}	DSICLK Cycle time		1.53	-	25	ns
T _{DSIR}	DSI Data Transfer Rate		80	-	1300	Mbps
T _{SETUP}	Data to Clock Setup time	≦ 1000Mbps	0.15	-	-	UI
			0.15	-	-	ns
		≧ 1000Mbps	0.2	-	-	UI
			0.15	-	-	ns
T _{HOLD}	Data to Clock Hold time	≦ 1000Mbps	0.15	-	-	UI
			0.15	-	-	ns
		≧ 1000Mbps	0.2	-	-	UI
			0.15	-	-	ns

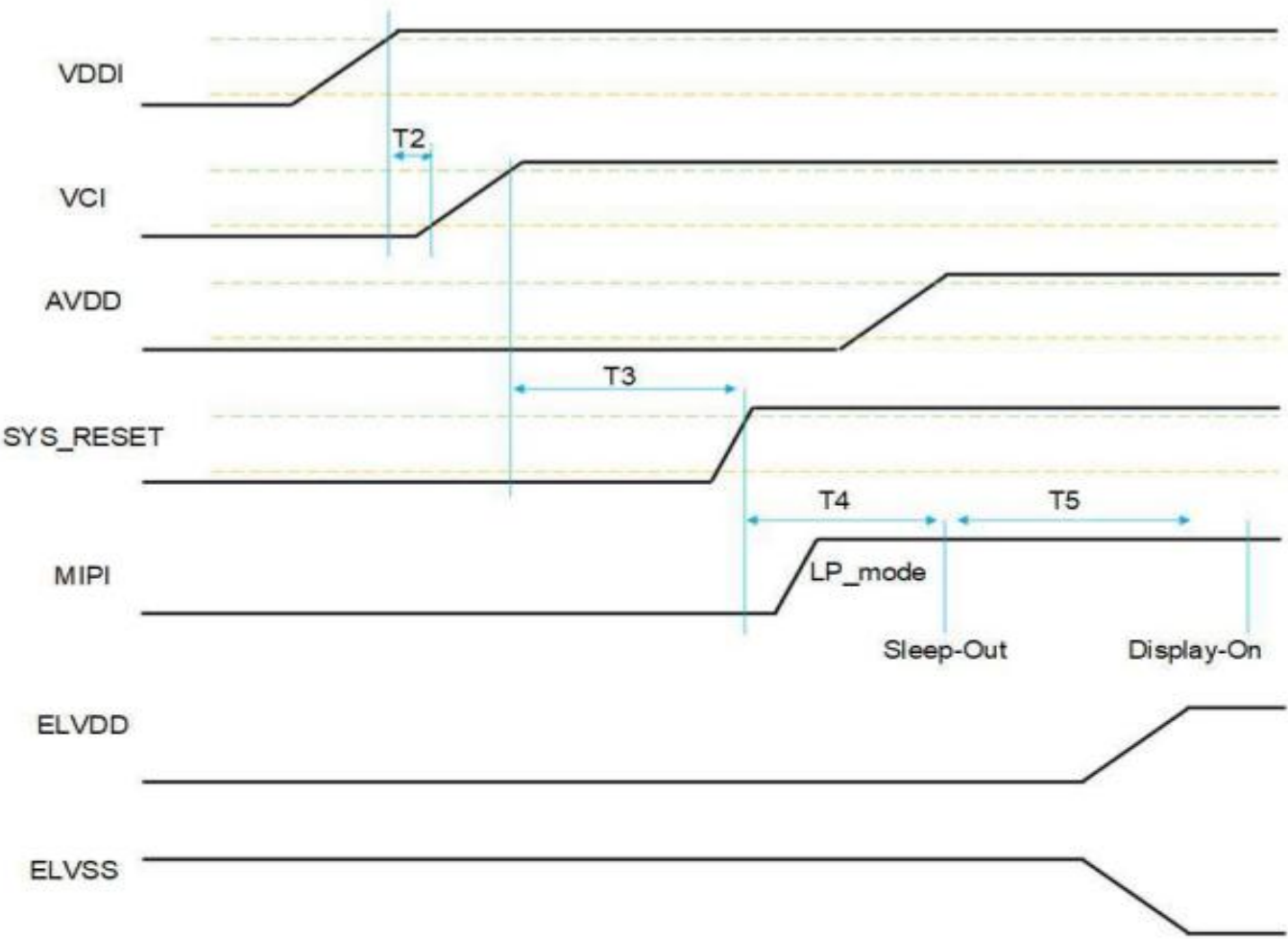
7.3 Global Operation Timings

Parameter	Description	Spec.			Unit
		Min.	Typ.	Max.	
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to beginning of $T_{CLK-TRAIL}$.	$60ns+52*U$ 1	-	-	-
$T_{CLK-PRE}$	Time that the HS clock shall be driver by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8	-	-	UI
$T_{CLK-PREPARE}$	Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-00 Line state starting the HS transmission.	38	-	95	ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the HS-00 state after the last payload clock bit of a HSHS transmission burst.	60	-	-	ns

$T_{CLK-PREPARE} + T_{CLK-ZERO}$	$T_{CLK-PREPARE}$ + Time to that the transmitter drives the HS-00 state prior to starting the clock.	300	-	-	ns
T_{EOT}	Transmitted time interval from the start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$, to the start of the LP-11 state following a HS burst.	-	-	$105ns + n \cdot 12 \cdot UI$	-
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following HS burst.	100	-	-	ns
$T_{HS-PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-00 Line state starting the HS transmission.	$40ns + 4 \cdot UI$	-	$85ns + 6 \cdot UI$	-
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + Time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	$145ns + 10 \cdot UI$	-	-	-
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst	$(n \cdot 8 \cdot UI, 60ns + n \cdot 4 \cdot UI)$ 中的最大值	-	-	-
T_{LPX}	Transmitted length of any Low-Power state period	50	-	-	ns
Ratio T_{LPX}	Ratio of $T_{LPX(MASTER)}/T_{LPX(SLAVE)}$ between Master and Slave side	2/3	-	3/2	-
T_{TA-GET}	Time that the new transmitter drives the bridge state (LP-00) after accepting control during a Link Turnaround.	$5 \cdot T_{LPX}$			-
T_{TA-GO}	Time that the new transmitter drives the bridge state (LP-00) before releasing control during a Link Turnaround.	$4 \cdot T_{LPX}$			-
$T_{TA-SURE}$	Time that the new transmitter waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link turnaround.	T_{LPX}	-	$2 \cdot T_{LPX}$	-

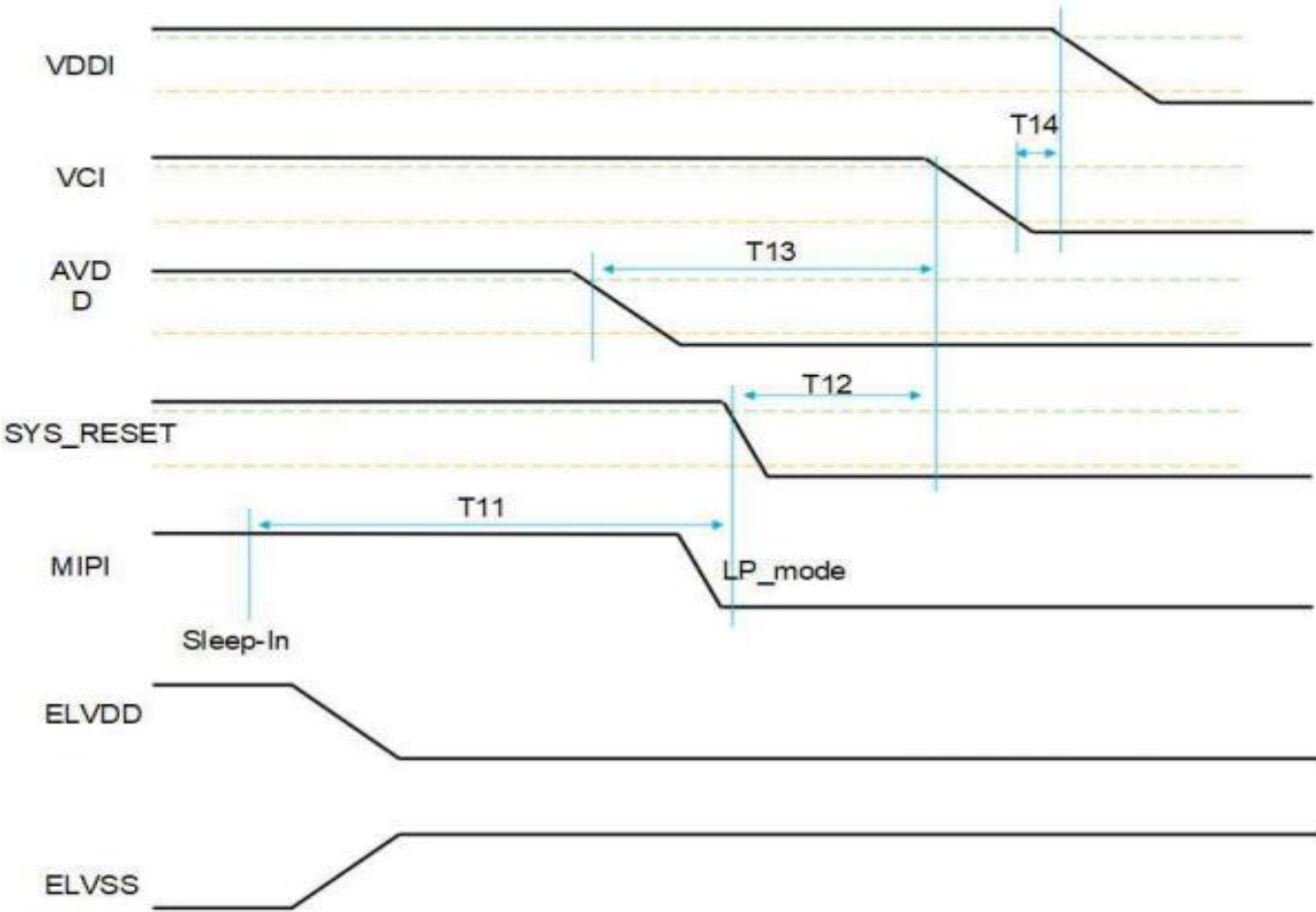
7.4 Recommended Operation Sequence

7.4.1 Display Power on Sequence



上电时序参数		Min.	Typ.	Max.	Unit
T2	VDDI 上电时间到 Reset 开始时间	50	-	-	us
T3	VCI 上电时间到 Reset 开始时间	6	-	-	ms
T4	复位完成到能响应 sleep-out 时间	36	--	-	ms
T5	从 sleep-out 到 display-on 时间	40	-	-	ms

7.4.2 Display Power off Sequence



下电时序参数		Min.	Typ.	Max.	Unit
T11	Sleepiin 到 AP 开始 Reset 的时间	45	-	-	ms
T12	Reset 到开始关掉 VCI 的时间	50	-	-	ms
T13	AVDD 下电完成到 VCI 开始下电的时间	50	-	-	ms
T14	VCI 下电完成到 VDDIO 开始下电的时间	50	-	-	ms

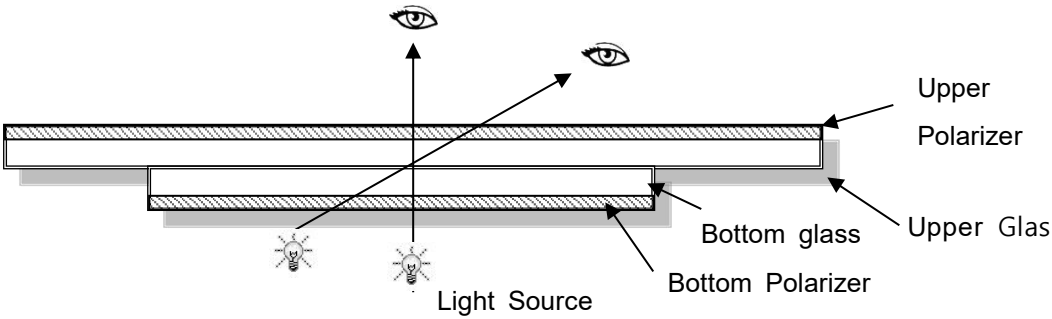
8. OLED Module Out-Going Quality Level

8.1 VISUAL & FUNCTION INSPECTION STANDARD

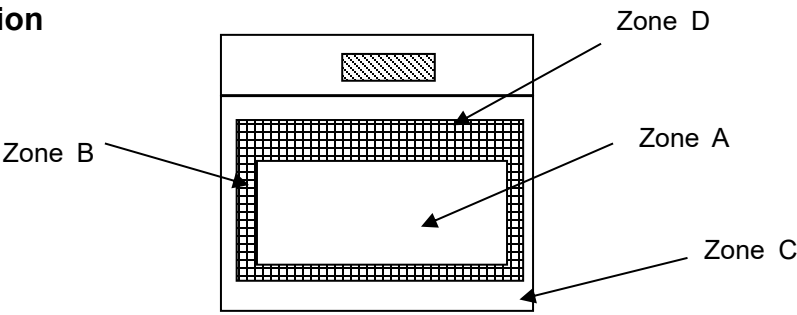
8.1.1 Inspection conditions

Inspection performed under the following conditions is recommended.

- Temperature : 25±5℃
- Humidity : 65%±10%RH
- Viewing Angle : Normal viewing Angle.
- Illumination: Single fluorescent lamp (300 to 700Lux)
- Viewing distance:30-50cm



8.1.2 Definition



- Zone A : Effective Viewing Area(Character or Digit can be seen)
- Zone B : Viewing Area except Zone A
- Zone C Cover (Zone A+Zone B) which can not be seen after assembly by customer .)
- Zone D : IC Bonding Area

Note:
 As a general rule ,visual defects in Zone C can be ignored when it doesn't effect product function or appearance after assembly by customer

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8.1.3 Sampling Plan

According to GB/T 2828.1-2012 ; , normal inspection, Class II

AQL:

Major defect	Minor defect
0.65	1.5

OLED: Liquid Crystal Display , LCM: Liquid Crystal Module

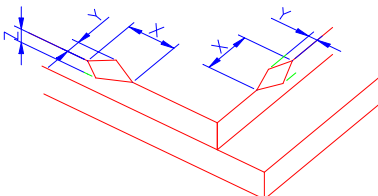
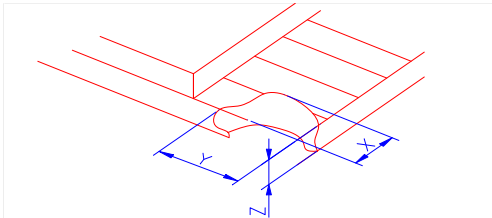
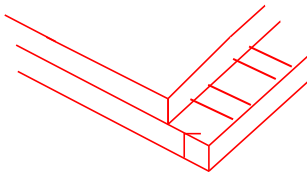
No	Items to be inspected	Criteria	Classification of defects
1	Functional defects	1) No display, Open or miss line 2) Display abnormally, Short 3) Backlight no lighting, abnormal lighting. etc	Major
2	Missing	Missing components and etc	
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed, deformation and etc	
4	Color tone	Color unevenness, refer to limited sample	Minor
5	Spot/Line defect	Light dot, Dim spot, (Note1) Polarizer Air Bubble, Polarizer accidented spot and etc	
6	Soldering appearance	Good soldering , Peeling off is not allowed and etc	
7	OLED/Polarizer	Black/White spot/line, scratch, crack, etc.	

Note1: a) Light dot: Dots appear bright and unchanged in size in which OLED panel is displaying under black pattern.

b) Dim dot: Dots appear dark and unchanged in size in which OLED panel is displaying under pure red, green, blue picture.

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8.1.4 Criteria (Visual)

Number	Items	Criteria(mm)						
1.0 OLED Crack/Broken NOTE: X: Length Y: Width Z: Height L: Length of IT O, T: Height of OLE D	(1) The edge of OLE D broken	 <table><tr><td>X</td><td>Y</td><td>Z</td></tr><tr><td>≤3.0mm</td><td><Inner border line of the seal</td><td>≤T</td></tr></table>	X	Y	Z	≤3.0mm	<Inner border line of the seal	≤T
	X	Y	Z					
	≤3.0mm	<Inner border line of the seal	≤T					
(2) OLED corner broken	 <table><tr><td>X</td><td>Y</td><td>Z</td></tr><tr><td>≤3.0mm</td><td>≤L</td><td>≤T</td></tr></table>	X	Y	Z	≤3.0mm	≤L	≤T	
X	Y	Z						
≤3.0mm	≤L	≤T						
(3) OLED crack	 Crack Not allowed							

Spot defect

2.0

Φ=(X+Y)/2

① light dot (black/white spot , pinhole, stain, etc.)

Zone Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.15	Ignore		Ignore
0.15<Φ≤0.25	3(distance ≥ 10mm)		
0.25<Φ≤0.4	2(distance ≥ 10mm)		
Φ>0.4	0		

② Dim spot (light leakage、dent、dark spot, etc)

Zone Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.15	Ignore		Ignore
0.15<Φ≤0.25	3(distance ≥ 10mm)		
0.25<Φ≤0.4	2(distance ≥ 10mm)		
Φ>0.4	0		

③ Polarizer accidented spot

Zone Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.2	Ignore		Ignore
0.2<Φ≤0.5	2(distance ≥ 10mm)		
Φ>0.5	0		

④Polarizer Bubble

Zone Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.2	Ignore		Ignore
0.2<Φ≤0.4	2(distance ≥ 10mm)		
0.4<Φ≤0.5	1		
Φ>0.5	0		



Note:

A) Bright dot: Dots appear bright and unchanged in size in which OLED panel is displaying under black pattern.

B) Dark dot: Dots appear dark and unchanged in size in which OLED panel is displaying under pure red, green, blue picture.

C) 2 dot adjacent = 1 pair = 2 dots

Picture:

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4.0	Line defect (OLED /Polarizer backlight black/white line, scratch, stain)  W: width, L : length N : Count	Width(mm)	Length(m m)	Acceptable Qty		
				A	B	C
		$\Phi \leq 0.05$	Ignore	Ignore		Ignore
		$0.05 < W \leq 0.06$	$L \leq 5.0$	$N \leq 3$		
		$0.06 < W \leq 0.08$	$L \leq 4.0$	$N \leq 2$		
		$W > 0.08$	Define as spot defect			

5.0	Electronic Components SMT.	Not allow missing parts, solderless connection, cold solder joint, mismatch, The positive and negative polarity opposite
6.0	Display color& Brightness.	1. Color: Measuring the color coordinates, The measurement standard according to the datasheet or samples. 2. Brightness: Measuring the brightness of White screen, The measurement standard according to the datasheet or Samples.
7.0	OLED Mura/Waving/ Hot spot	Not visible through 5% ND filter in 50% gray or judge by limit sample if necessary.

Criteria (functional items)

Number	Items	Criteria (mm)
1	No display	Not allowed
2	Missing segment	Not allowed
3	Short	Not allowed
4	Backlight no lighting	Not allowed
5	CTP no function	Not allowed

9. Reliability Test Result

Item	Condition	Inspection after test
High Temperature Operating	85°C, 96H	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the OLED; 2.Non-display; 3.Missing segments/line; 4.Glass crack; 5.Current IDD is twice higher than initial value.
Low Temperature Operating	-30°C, 96HR	
High Temperature Storage	85°C, 96HR	
Low Temperature Storage	-40°C, 96HR	
High Temperature & High Operating	+60°C, 90% RH ,96 hours.	
Thermal Shock (Non-operation)	-40°C, 30 min ↔ 80°C, 30 min, Change time: 5min 20CYC.	
ESD test	C=150pF, R=330, 5points/panel Air: ±8KV, 5times; Contact: ±10KV, 5 times; (Environment: 15°C~35°C, 30%~60%).	
Vibration (Non-operation)	Frequency range: 10~55Hz, Stroke: 1.5mm Sweep: 10Hz~55Hz~10Hz 2 hours for each direction of X.Y.Z. (6 hours for total) (Package condition).	
Box Drop Test	1 Corner 3 Edges 6 faces, 80cm (MEDIUM BOX)	

Remark:

- 1.The test samples should be applied to only one test item.
- 2.Sample size for each test item is 5~10pcs.
- 3.For Damp Proof Test, Pure water(Resistance > 10MΩ) should be used.
- 4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.
- 5.Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

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10. Cautions and Handling Precautions

10.1 Handling and Operating the Module

(1) When the module is assembled, it should be attached to the system firmly.

Do not warp or twist the module during assembly work.

(2) Protect the module from physical shock or any force. In addition to damage, this may cause improper operation or damage to the module and back-light unit.

(3) Note that polarizer is very fragile and could be easily damaged. Do not press or scratch the surface.

(4) Do not allow drops of water or chemicals to remain on the display surface.

If you have the droplets for a long time, staining and discoloration may occur.

(5) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.

(6) The desirable cleaners are water, IPA (Isopropyl Alcohol) or Hexane.

Do not use ketene type materials (ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanent damage to the polarizer due to chemical reaction.

(7) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth. In case of contact with hands, legs, or clothes, it must be washed away thoroughly with soap.

(8) Protect the module from static; it may cause damage to the CMOS ICs.

(9) Use finger-stalls with soft gloves in order to keep display clean during the incoming inspection and assembly process.

(10) Do not disassemble the module.

(11) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.

(12) Pins of I/F connector shall not be touched directly with bare hands.

(13) Do not connect, disconnect the module in the "Power ON" condition.

(14) Power supply should always be turned on/off by the item 6.1 Power On Sequence & 6.2 Power Off Sequence

10.2 Storage and Transportation.

(1) Do not leave the panel in high temperature, and high humidity for a long time.

It is highly recommended to store the module with temperature from 0 to 35 °C and relative humidity of less than 70%

(2) Do not store the TFT-OLED module in direct sunlight.

(3) The module shall be stored in a dark place. When storing the modules for a long time, be sure to adopt effective measures for protecting the modules from strong ultraviolet radiation, sunlight, or fluorescent light.

(4) It is recommended that the modules should be stored under a condition where no condensation is allowed. Formation of dewdrops may cause an abnormal operation or a failure of the module.

In particular, the greatest possible care should be taken to prevent any module from being operated where condensation has occurred inside.

(5) This panel has its circuitry FPC on the bottom side and should be handled carefully in order not to be stressed.

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11. Packing

----TBD-----

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